History dependence without qualified common support: Limits of learned native bitrate control

networkRLCD project
Retrospective research-decision synthesis

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Abstract

Fine-grained learned bitrate control requires a defensible connection between sender information, randomized action effects and future delivery risk. We synthesize frozen native Chromium and simulator studies that did not establish that connection robustly. Two bounded native causal findings anchor the result: randomized early service history changed later bandwidth estimates (BWE), and a fixed BWE-threshold policy reduced timely delivery by 14.1 percentage points (unadjusted 95% interval $[-22.6, -5.5]$) relative to fixed 0.65. A positive secondary quality interaction in a local action study did not receive an adequately covered independent replication. Subsequent support studies did not establish reproducible conditional native state. History effects varied with content and time; no registered sustained convergence was found by 36 s. Available mechanism channels failed their discrimination criteria. Internal transport events existed, but their online freshness and identity did not jointly qualify; a proposed instrumented fixture was never built because resource admission failed. Separately, an 816-episode simulator screen failed robust out-of-family prediction and action discrimination. Interval coverage met its registered minimum, but excessive width and failed probability calibration prevented useful supported uncertainty. Its native factual panel remains unopened. These findings support RD1's stopping decision under the available fixture/data assumptions. They do not establish universal control impossibility, permanent history memory, or the irrelevance of observable history. No successor, additional model or controller is introduced.

1 Research question and contribution

Interactive media must balance source quality, timely delivery and network cost; these goals are not interchangeable [2]. A native sender's reported bandwidth estimate (BWE) is an output of a transport/encoder feedback system, not an exogenous measurement of usable capacity. A controller could exploit observable history only if its action-value and risk claims are identifiable for the relevant population and survive validation. Good mean prediction in one regime is insufficient; useful uncertainty must also transport [3].

The original networkRLCD program investigated calibrated learned adaptation in simulation, then native action effects and causal support. Its evidence anchor is commit `f109ab2`, the completed PO1 result. Its scientific stopping decision is RD1 at `050e4f4` [1]; this publication pass changes neither evidence nor decision. All prior artifacts, including the earlier controller-centric manuscript and the RD1 edition, are preserved.

The central contribution is an empirically delimited failure to connect local causal effects to reproducible conditional control value. Bounded causal history effects and local policy harm were observed. Reproducible conditional native state, internal causal observability, robust synthetic action discrimination and useful calibrated uncertainty were not established. Each failed link has its own experimental unit, gate and scope. This separation preserves what was learned without treating a support failure as an action outcome or inaccessible state as an irrelevant mechanism. The result is a stopping decision under explicit assumptions, not a controller or safety certificate.

Earlier simulator studies retain their reported local successes and failures in the frozen prior edition [6]. They used different representations, policies and targets; PO1 does not erase those results or supply their native validation. Passing some synthetic population-panel criteria did not establish uniform within-family calibration or native control value.

2 Estimands and what failure means

Let $\mathcal{I}_t$ contain sender observations, past commands and records *actually received* by decision time t. Let U_t denote unobserved transport state, A_t a bitrate action and Y a specified future endpoint. Three questions must remain separate:

$$\tau_H(t; S) = \mathbb{E}[X_t(H_d, S) - X_t(H_a, S)], \tag{1}$$

$$\tau_A(s; \mu) = \mathbb{E}_\mu[Y(a_1) - Y(a_0) \mid g(\mathcal{I}_t) = s], \tag{2}$$

$$V_\mu(\pi) - V_\mu(\pi_0) = \mathbb{E}_\mu[Y(\pi) - Y(\pi_0)]. \tag{3}$$

The first is a total effect of randomized early service order H followed by common input service S. For M5, X_t is the mean log2-BWE response over the registered eleven observations, with the fixed normalization specified below. The second is a conditional action effect under a specified history/episode distribution μ. The third is a complete policy or sequence contrast. A factual predictor $\mathbb{E}[B_{t+1:t+h} \mid \mathcal{I}_t, \text{logging regime}]$ is a fourth, predictive target; it is not a native counterfactual action value.

Randomized actions can identify local or marginal effects despite hidden U_t, subject to consistency, actual randomization/positivity and the stated population. Observing every GCC mechanism or finding a stationary Markov state is not a universal prerequisite for randomized policy inference. Native V9 and V10 demonstrate that bounded causal contrasts can be estimated in this fixture. The project's intended *transferable fine-grained conditional value*, however, additionally needs support and a defensible relationship between the selected observable context and its future response across histories and sources. That relationship has not qualified. Changing μ can change the latent-history mixture even when instantaneous BWE is the same [1].

We distinguish a supported bounded causal effect, a contradicted specific superiority claim, an unsuccessful registered recipe, an unestimable comparison after a gate failure, a fixture/practical limit, and an unresolved mechanism. None alone is a theorem that all learned native control is impossible. Failed support sessions are not treatment outcomes; zero emitted predictions have unestimable precision. Receiver outcomes left unopened in support studies are not assigned zero effects.

3 Evidence handling, sources and technical controls

This is a synthesis of committed result reports and their pinned provenance, not a meta-analysis or raw-data reanalysis. No captures are restored, no closed validation representation is searched, no receiver outcome is opened and no new model is fitted. RD1's evidence ledger binds cited reports, protocols and selected storage receipts to SHA-256. The publication claim map adds exact source locators, classifications and independence units. A preservation audit checks every file tracked at the publication baseline, including RD1. This edition is separate so that old scientific freezes remain valid.

Native Chromium outcomes, native responses to controlled synthetic service, measured-rate replay and simulated outcomes are distinct. Native V9/V10 used generated content and a controlled schedule; their native telemetry is not a fluid simulation. V11D crossed licensed source content with four strata nested in two measured campaigns. V12 used three campaigns. V13's LTE/EVDO bundles come from one measurement program. M5 onward reuse licensed content and measured-source apparatus; many later common-service plateaus are controlled inputs. A fresh browser process is a fresh run, not a new independent content production or network campaign. Measured application-throughput replay is not a live Internet trial or an observation of every end-to-end queue [7, 8, 9, 10, 11, 12, 18, 19].

The managed browser policy prohibits nonproxied UDP. Qualified native acquisition used TURN/TCP without changing that policy. Sender/relay clock markers, actual-run held-out clock coverage, native initialization, cap/jitter readback, replay fidelity and cleanup were enforced before admitted analyses. Later receipt-based features could not use future clock anchors or backdated event timestamps. Technical qualification is necessary for attribution, but is not scientific common support. Original failures and prospective repairs remain separate studies [7, 8, 12, 13, 23].

4 Native action evidence did not establish transferable value

4.1 A local secondary signal and a harmful policy

Ratios 0.65 and 1.05 denote multiples of pre-action sender BWE. Native V9 randomized these BWE-relative caps at six 3 s boundaries in 48 sessions, nested in 12 generated-scene seed clusters. Its high/low BWE split

was prospectively fixed at 500 kbps. Eleven clusters met the registered state/action counts; one remained in the evidence but outside the complete-cell analysis. The primary high-minus-low BWE interaction in the 150 ms on-time fraction was -1.8 percentage points, with an unadjusted 95% t interval $[-29.4, 25.8]$. The secondary PSNR interaction was 0.508 dB $[0.077, 0.938]$ [7]. Both intervals use 11 seed-cluster contrasts (10 degrees of freedom); holds and frames are not independent replicates. PSNR is sampled among on-time frames, not common-frame quality under all actions or human QoE. This supports a local secondary signal under that logging regime, not a stable state value or policy benefit. The primary delivery interaction remains unresolved.

Native V10 tested the direct whole-session policy implication: use 1.05 at or above the frozen 500 kbps BWE boundary and 0.65 below, versus fixed 0.65. All 48 sessions qualified; two repeats/policy were averaged within each of 12 scene seeds. The threshold policy's on-time contrast was -14.1 percentage points $[-22.6, -5.5]$. PSNR was $+0.257$ dB $[-0.132, 0.646]$. Policy assignment was for a whole session; the delivery endpoint was the 150 ms on-time frame fraction over six registered 800 ms post-acknowledgment windows. Both intervals are unadjusted 95% paired t intervals over 12 scene-seed contrasts (11 degrees of freedom), not 48 independent source units [8]. The specific threshold policy's local superiority claim is contradicted. This policy is closed and is not retuned. A positive local quality interaction did not translate into whole-session policy benefit.

4.2 Under-covered replication and action stages that never opened

V11D completed 64 qualified native captures from four content groups, four network-family strata and two measured campaigns. Only 9/16 crossed cells had at least two holds in every state/action bucket; at least 13 were required. Only seven cells had PSNR in every bucket, also below the required 13 primary-estimable cells. Their descriptive PSNR interaction was -0.609 dB, with campaign means -0.419 dB (NordicDat) and -0.862 dB (Rome), from four and three primary-estimable cells respectively [9]. Both directions oppose V9, but the coverage gate failed. This is a negative, under-covered replication, not a population reversal estimate. No missing bucket is imputed; 16 crossed cells and four nested families are not independent campaigns.

V12 attempted all 24 support originals before randomizing any action. Twenty-three ordinary captures sealed and one preserved capture failed native media statistics. Ten passed all gates; none of six cells had complete frozen low/high-BWE support. The 23 ordinary captures passed clocks, transport, replay and cleanup. All 48 planned action sessions stayed unrun [10]. The six cells are two content groups $\times$ three campaigns, with two support checks per state per cell. Capacity-based source hints did not guarantee native estimator occupancy. The failed original is not counted as technically qualified.

V13 replaced the binary split with a preregistered continuous BWE interaction, but still required support before actions. All 32 originals individually qualified. Six of 16 content/profile contexts exceeded the 0.25-octave fresh-repeat limit; only one of four cells passed completely. All cells passed profile range and SD, yet the cross-cell support intersection was empty. The 16 contexts are two contents $\times$ eight profiles, with two repeats each; the four cells are content $\times$ replay family. Its 64 randomized action sessions remain unrun [11]. V12/V13 supply no action-effect or QoE result. Individual technical qualification cannot substitute for panel support.

5 History is causal; usable common state did not qualify

5.1 M5's supported finite-history effect

M5 qualified 84 original sessions and 42 randomized prefix pairs before its causal comparison. Within each of two reused contents, 21 adjacent pairs received ascending/descending permutations of the same early service multiset and then the same measured-source suffix. Actual-run clocks and suffix-exposure checks passed [12].

At 18 s, descending-minus-ascending mean log2-BWE differences were 0.20746 octave $[0.09910, 0.31413]$ for Kampar and 0.14750 $[0.09807, 0.19353]$ for Volga. Both registered distribution tests reject at local $\alpha = 0.025$. The endpoint is an eleven-sample mean of $\log_2(\text{BWE}/500000)$; normalization is not a binary threshold. Brackets are 95% paired-bootstrap intervals from the frozen 10,000-draw procedure, using 21 randomized pairs per content. The exact paired distribution tests and mean intervals answer different questions. Contents and studies are not pooled; intervals describe repeated fixed-recipe capture uncertainty, not a new-source population [12].

Later signed effects changed direction and neither content satisfied sustained convergence through 36 s. Thus early service history causally changes later native BWE in this fixed recipe. This does not establish permanent memory, behavior beyond the horizon, independent network replication, or a unique mediator. Holding later service input common does not force identical internal queue, encoder or estimator trajectories. These are two reused contents and one reused LTE measurement source [12].

5.2 Matching, conditioning and dynamic-state failures

M6R qualified 32 originals but only four of six frozen content/landmark strata met membership coverage. Adding recent encode-time history excluded zero BWE-only matches; the selected state had exactly the baseline membership. Conditional future comparisons were therefore not run [13]. The six strata are two contents × three landmarks; eight originals/history/content share those landmarks. This is sparse overlap and absent matching discrimination at the frozen calipers, not a null causal history effect. Conditioning on post-prefix BWE/history cannot by itself identify a direct prefix effect or causal mediation.

M7RA's four feasibility originals passed technical gates, but only two reached the instantaneous target and one met sustained-history membership. Neither region qualified across histories. Its 128-session validation remains untouched [17]. The four contexts use one reused content, two targets and two histories, with one original per context. These are deterministic gate checks, not population rates; the larger validation requirements were never assessed. A fixed conditioning recipe did not create common state.

M8's 16 characterization originals under repeated 24 s controlled-service plateaus showed strongly asymmetric native dynamics: rise t50 about 7.4–10.2 s and t90 13.3–15.9 s, versus fall t50 about 0.39–0.79 s. These are envelopes of input-normalized onset intervals across observed contexts, including sampling and input-clock uncertainty, not population confidence intervals or time to an assumed equilibrium. The 16 runs cross two reused contents, two prefixes, two initial orders and two repeats; two earlier instrumentation originals qualified separately [18]. These are native content/encoder/transport responses to controlled service, not simulated BWE or isolated GCC parameters.

At each frozen candidate age (6, 9, 12 and 15 s), none of 64 plateau occurrences was future-verified under the D1 local-settling rule. The occurrences are four plateaus nested in each of 16 native runs. D1 emitted some tags, but none verified; no conditioning age was selected [18]. The onset search is censored and cannot prove permanent nonsettling.

M9R prospectively extended observation to a 64 s common plateau; all eight instrumentation and 64 validation originals qualified. Instantaneous BWE (B0) tagged 64 originals and D1 tagged 55, with no future-verified first predictions. The phase-aware P2 rule emitted no tag, making its positive precision unestimable. Full four-pair matching coverage fell from 42/64 strata for BWE-only to 13/64 for the compact history vector. Residual future divergence remained in some matched contexts [19]. Validation crosses two reused contents, four service inputs, two histories and four repeats/history. The 64 matching strata cross two contents, four inputs and eight landmarks; landmarks share runs. There were zero future-stability labels in 3,200 nested run/grid queries, not 3,200 independent experiments. Longer observation did not establish prospectively useful state; it does not license another wait or classifier on these closed captures.

6 Mechanism observability and build practicality

6.1 M10R: available counters did not resolve residual divergence

M10R qualified four instrumentation and eight fresh development originals at 1.414 Mbps common input. It observed acknowledged owned-relay queue/service counters, native RTCP RTT, RTT-counter detection age, send/cap history and encoder behavior. No channel met the frozen repeated same-direction contrast with controls in both contents. Of 96 nested pair/landmark contrasts, 49 compact-matched; 23 of those 49 exceeded the 0.25-octave future-distance bound. The eight development runs are two reused contents × two histories × two repeats. Repeated landmarks and prepaired/crossed/within-history contrasts share those runs. There were 0/400 positive future-stationary labels, nested in eight originals; no stability classifier was fitted [21].

RTT-counter detection age is not transport-CC generation/arrival age; the owned relay is one queue, not every queue in the TURN/TCP path. Available contrasts may be consequences of BWE or shared causes. Seven of 14 compact-matched within-history comparisons also diverged, so residual variability is not all an isolated prefix effect. The assay established neither causal mediation nor a stationary or prospectively reproducible trajectory state. Some repeatable moving marginal distributions are insufficient [21].

6.2 M11R: real events lacked jointly useful freshness and identity

Both M11R native capability originals qualified ordinary acquisition. Identified RTC logs contained actual TCC, delay/loss estimator-update and probe events. The gate required at least 90% of each event family sender-available within 250 ms in each original, clock uncertainty at most 50 ms, no continuous-feedback delivery gap above 1,000

ms and stable PC/SSRC/controller attribution. Fresh TCC counts were 31/1,739 (1.78%) and 33/1,747 (1.89%); fresh BWE-update counts 30/2,110 (1.42%) and 37/2,090 (1.77%); fresh probe counts 0/16 and 0/31. Events are nested in two capability originals, not independent trials. First identified export was about 59 s after measurement start, followed by roughly 5 s batches [23].

Verbose target-update lines were fresh but lacked controller identity; a target emission is not every estimator update. Trace exports supplied no online sender receipt. No relay/socket/RTT/RTP-burst equivalent proxy qualified. All required channels qualified in 0/2 originals. A record becomes causally usable at its receipt, not its earlier producer timestamp. Full internal pacer queue/credit/cwnd and probe-active state were not obtained. Transport-timing explanations of residual divergence remain untested in this fixture, not disproved [23].

6.3 IF1: the instrumented alternative was not built

IF1 specified pinned Chromium/WebRTC hooks, stable identity, nonblocking event publication and sender-clock receipt. Its formal admission audit found 17.7 GB free disk on a 33.8 GB filesystem, below the source's 100 GB minimum and IF1's 130 GB/16 GiB/4-core build requirement; no resource-attested remote builder or provenance-qualified artifact was available. There were zero compiler attempts, C++ instrumentation patches, full source checkouts or new native sessions [25]. This is a practical fixture-admission failure. Neither live instrumentability nor instrumented Chromium's scientific informativeness was tested. Static hook discovery is not channel qualification, common support or action value.

7 PO1: three distinct synthetic failures

PO1 was preregistered as a necessary-condition screen before a fresh native factual panel. It uses the existing fluid bottleneck, simplified AckedBwe and assumed encoder response, not production libwebrtc [5, 26]. Three fits use 384 fitting, 144 calibration and 288 held-out validation episodes in total; validation covers four generator families disjoint from fitting and calibration. All 816 episodes and all binary training/calibration support gates qualified. Models/calibrators were locked before validation decoding. There are 24 validation episodes/family/fit, hence 72/family across three separately fitted seed batches, all within one simulator implementation. Six decisions/episode are nested; uncertainty resamples episodes within fit, not holds.

The fixed 1.05 synthetic logger is distinct from frozen native fixed 0.65. At each decision, features are saved before hindsight branches compare 0.65 for 5 s with 1.05 for 3 s then 0.65 for 2 s. A separate factual branch continues logging 1.05 for 5 s. The factual regression target is next-5-s mean log2-BWE minus current log2-BWE. Synthetic advantage is mean 5 s utility of the high-then-low branch minus the always-low branch. Binary targets are a BWE drop of at least 0.5 octave, advantage above 0.02 utility, and high-branch unsafe fraction above 0.10, respectively [26]. Only the simulator supplies these counterfactual labels; none is transferred to native data or human QoE.

S1 uses current values/masks and past 3 s means, slopes and missing fractions of BWE, delivered throughput, RTT, requested cap and actual send rate. H1 adds only ordered 1/2/3 s samples/masks. These are fixed-penalty linear/logistic models; there is no recurrent model or learned policy. Baselines include persistence, trailing mean, clipped trend, train-only constant and instantaneous linear models as well as S1. Calibration is held out; proper prediction scores are not themselves control guarantees [4, 3, 26].

	Factual BWE MAE (octaves)			Synthetic advantage MAE		
Family	Persistence	S1	H1	Constant	S1	H1
sawtooth	0.740	0.457	0.469	0.256	0.204	0.203
square_wave	0.684	0.413	0.421	0.251	0.200	0.199
slow_fade	0.587	0.447	0.469	0.183	0.152	0.154
feedback_flap	0.407	0.716	0.724	0.080	0.221	0.225

Table 1: PO1 frozen validation values: 72 episodes/family across three fixed fits, six nested decisions/episode. S1 is the fixed-history summary baseline; H1 adds ordered lags. Lower is better. These are simulator results, not native effects.

Prediction did not transport robustly. H1 improves on persistence/current-value baselines in three families, but its BWE MAE is worse than S1 in every family. Under feedback-flap H1 error is 0.724 versus persistence 0.407, 77.8% worse; S1 also fails there. This is evidence against this package's robust out-of-family incremental information, not all history. The assay tests a representation/model/logging/calibration package, not the information-theoretic sufficiency of every possible function of history [26]. That distinction does not waive the registered stop on complexity escalation.

Action discrimination did not qualify. H1 advantage-MAE gains over S1 were only 0.19% and 0.43% in two families, below the frozen 5% requirement, with lower bootstrap improvement bounds below zero and fit-level sign reversals. The other families worsened. These are the registered within-fit episode-bootstrap 5th–95th percentile bounds, conditional on three fixed fits, not uncertainty over simulator implementations or network populations. Feedback-flap H1 advantage MAE 0.225 versus constant 0.080 is 179.4% worse [26]. These are synthetic action-value prediction errors, not native causal effects or an executed policy's reward. Oracle labels do not imply that observable history can discriminate actions under shift.

Coverage passed its minimum; useful uncertainty did not. Nominal 90% sequence intervals use calibration-episode maximum absolute residuals over six decisions. H1 BWE full widths are 4.08–5.40 octaves versus maximum 1.0; advantage widths 2.96–3.31 utility units versus maximum 0.50. Aggregate coverage met the registered 80% minimum and within-fit coverage was at least its 75% minimum, but excessive widths fail usefulness. This is not a claim of exact 90% coverage in every family, nor a distribution-free OOD guarantee. Every family fails the binary calibration conjunction; feedback-flap ECE is 0.397 for BWE drop and 0.321 for beneficial advantage. That family's zero unsafe-positive labels among 432 nested validation decisions (72 episodes $\times$ six decisions) cannot establish risk sensitivity [26].

An offline supported-action assertion requires an advantage lower bound greater than 0.02 utility, benefit probability at least 0.90 and unsafe probability at most 0.10. None qualified in any family, versus a minimum 30 asserted episodes/family and a Wilson upper episode-error bound at most 0.10. With zero assertions that upper bound is recorded as 1.0. Zero assertions are insufficient supported discrimination, not demonstrated safety. No learned action was executed. The registered stop leaves all 64 native factual sessions untouched; there is no PO1 native predictive result [26].

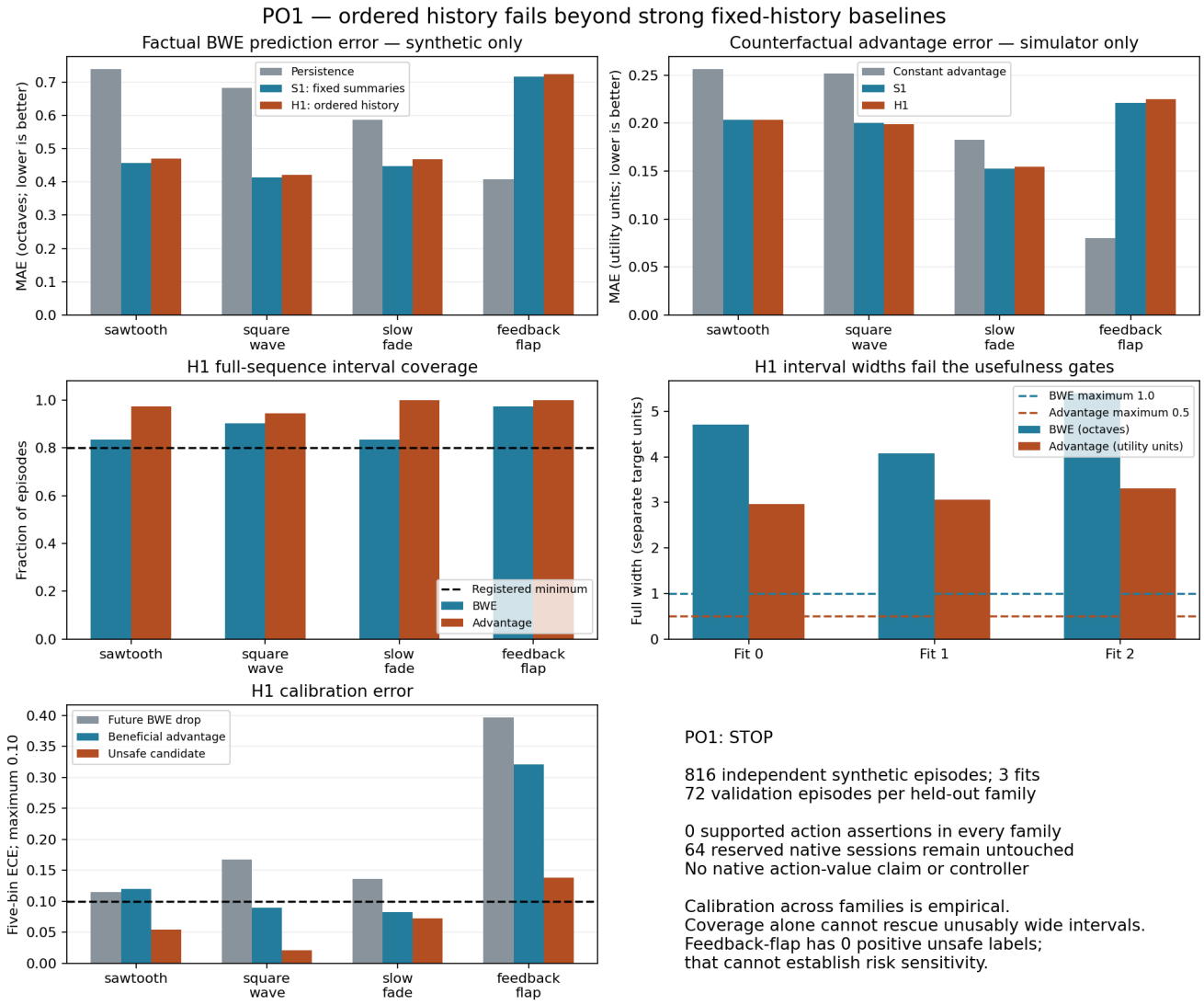


Figure 1: Frozen PO1 metrics from its committed report, with no new analysis. Baseline comparisons, calibration error and interval widths must be read alongside coverage. All families fail the preregistered conjunction. Families, fits and decisions are not independent simulator implementations or native replications.

8 Scientific stopping decision

The active fine-grained learned native-control objective is closed under the available fixture/data assumptions. No tested state passed its prospective support contract; the mechanism route reached an observability boundary; the new fixture was not admitted; PO1 did not supply robust predictive discrimination or useful uncertainty. These failures cannot be repaired by pooling sources, relaxing calipers, opening unused allocations, extending waits, increasing model complexity or searching closed validation. The PO1 stop remains exact: no history expansion, retraining with more complex sequence models, calibration/threshold tuning, native factual panel, controller fit or safe-policy efficacy work from that representation [1, 26].

The untested causal role of transport timing and scientific value of an unbuilt instrumented fixture remain unresolved. They limit the current conclusion; they are not evidence that another experiment would succeed. RD1's conditional discussion remains in its frozen decision document. This paper selects and proposes no successor. The publication outcome is the bounded negative result.

9 Limitations and nonclaims

These are adaptive research questions with prospectively frozen individual studies, not one preregistered omnibus experiment. Their sample counts cannot be summed into a population confidence claim. Generated scenes, reused clips, directions/windows, repeated landmarks and controlled inputs stay within their source/campaign clusters. PSNR and 150 ms frame-readback delivery are local proxies, not human QoE or live deployment safety. Technical qualification does not establish an empty end-to-end path queue or complete estimator observability.

Specific initial instrumentation failures remain failures: M6's 252 failed development-clock cutoff bounds (not 252 native sessions); M7's four scheduler originals, including 39/104 early service commands; M7R's four launcher attempts and zero native sessions; M9's two excessive-gap originals among 64 validation captures; M10's four cleanup overruns; and M11's two originals with SSRC0 configuration identity [14, 15, 16, 20, 22, 24]. Prospective repairs use fresh captures without changing numerical gates. None is recast as a scientific null or erased by a repair. IF1 was unattempted at build stage. P2 emitted no tags. Neither circumstance supplies evidence for a zero mechanism effect.

We have not proved permanent history memory, universal nonsettling, causal mediation by an internal queue, impossibility of instrumented Chromium, or a general inability to identify randomized native policy effects. Synthetic calibration failure is not native calibration failure; no PO1 native prediction was measured. Conversely, a bounded positive simulator or local native estimate does not establish broad native control value. The missing evidence is not replaced with a stronger claim.

10 Conclusion

Native early-service history has a qualified finite-horizon causal effect. Local randomized action and policy contrasts can be measured, including a harmful threshold policy. Yet the registered route to reproducible native conditional value did not qualify. Current diagnostics leave internal timing unobserved for causal online use; the instrumented build was never attempted; the compact synthetic history screen failed robust action discrimination and useful calibrated uncertainty under family shift. Together these results justify stopping the current fine-grained learned-controller objective under the available fixture and data assumptions. RD1 is the scientific stopping decision. All frozen evidence and untouched allocations remain intact. No reserved panel is opened, no successor is created and no model or controller is fitted or promoted.

A Reproducibility and provenance

RD1's machine-readable evidence ledger is `data/research-decision-rd1/evidence-ledger.json`; its prior-file freeze is `data/research-decision-rd1/prior-freeze-sha256.json`. RD1 sources, PDF and audit are preserved in a private immutable archive whose receipt records fresh object/member size and SHA-256 verification. This publication pass reads committed summaries, not raw native captures. It does not register an experiment.

The paragraph-level source map is `data/publication-readiness-rd1/claim-traceability.json`; verification is recorded in `reports/publication-readiness-rd1-audit.json`. All 1,698 files tracked at publication baseline `02c3693`, including RD1's decision, source edition, reports, receipts and PDF, remain byte-identical.

Publication sources and PDF are in `paper/negative-result-publication/`. All project reports cited here are pinned in the public RD1 source tree at `050e4f4`. Private raw access is restricted; committed reports and source locators support the numerical claims without implying that raw native data or licensed media are public.

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[12] networkRLCD. M5: qualified instrumentation makes the prefix-history effect estimable. `docs/NATIVE_PREFIX_MEMORY_M5_RESULTS.md`.

[13] networkRLCD. M6R: history-state common support did not qualify. `docs/NATIVE_HISTORY_STATE_M6R_RESULTS.md`.

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[15] networkRLCD. M7: instrumentation-limited feasibility failure. `docs/NATIVE_STATE_CONDITIONING_M7_RESULTS.md`.

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[17] networkRLCD. M7RA: qualified instrumentation, failed prospective state creation. `docs/NATIVE_STATE_CONDITIONING_M7RA_RESULTS.md`.

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[20] networkRLCD. M9: instrumentation-limited phase-state validation. `docs/NATIVE_PHASE_STATE_M9_RESULTS.md`.

[21] networkRLCD. M10R: fresh causal mechanism observability assay; qualified observability, mechanism discrimination not established. `docs/NATIVE_MECHANISM_OBSERVABILITY_M10R_RESULTS.md`; `docs/NATIVE_MECHANISM_OBSERVABILITY_M10R_INTERPRETATION.md`.

[22] networkRLCD. M10: cleanup instrumentation-limited stop. `docs/NATIVE_MECHANISM_OBSERVABILITY_M10_RESULTS.md`.

[23] networkRLCD. M11R: transport-control timing did not qualify for causal use. `docs/NATIVE_TRANSPORT_TIMING_M11R_RESULTS.md`.

[24] networkRLCD. M11: sender-identity instrumentation limitation. `docs/NATIVE_TRANSPORT_TIMING_M11_RESULTS.md`.

[25] networkRLCD. IF1: registered instrumented-fixture plan; build admission failed. `docs/NATIVE_INSTRUMENTED_FIXTURE_IF1_RESULT.md`.

[26] networkRLCD. PO1: bounded observable-history information screen; result: stop the bounded history-information route. Registration `d43d407`, result `f109ab2`. `docs/PARTIAL_OBSERVABILITY_PO1_PLAN.md`; `docs/PARTIAL_OBSERVABILITY_PO1_RESULTS.md`.